

Identify the vertex, focus, axis of symmetry, directrix, direction of opening, min/max value, length of the latus rectum, and the x- and y-intercepts of each.

7) $-2x^2 - 4x + y + 70 = 0$

8) $2y^2 + x + 20y + 51 = 0$

Use the information provided to write the transformational form equation of each parabola.

9) Vertex: $(-1, -3)$, Focus: $\left(-\frac{17}{16}, -3\right)$

10) Vertex: $(-3, 0)$, Focus: $\left(-\frac{47}{16}, 0\right)$

11) Vertex: $(-8, 5)$, Directrix: $y = \frac{19}{4}$

12) Opens left or right
Vertex: $(-7, 9)$
Passes through: $(-4, 8)$

13) Vertex: $(-2, -3)$, x-intercept: -11

14) Opens up or down, and passes
through $(-4, -3)$, $(-9, 27)$, and $(-3, 3)$